Fat soluble vitamins





Vitamin A, D, E & K





Vitamin A

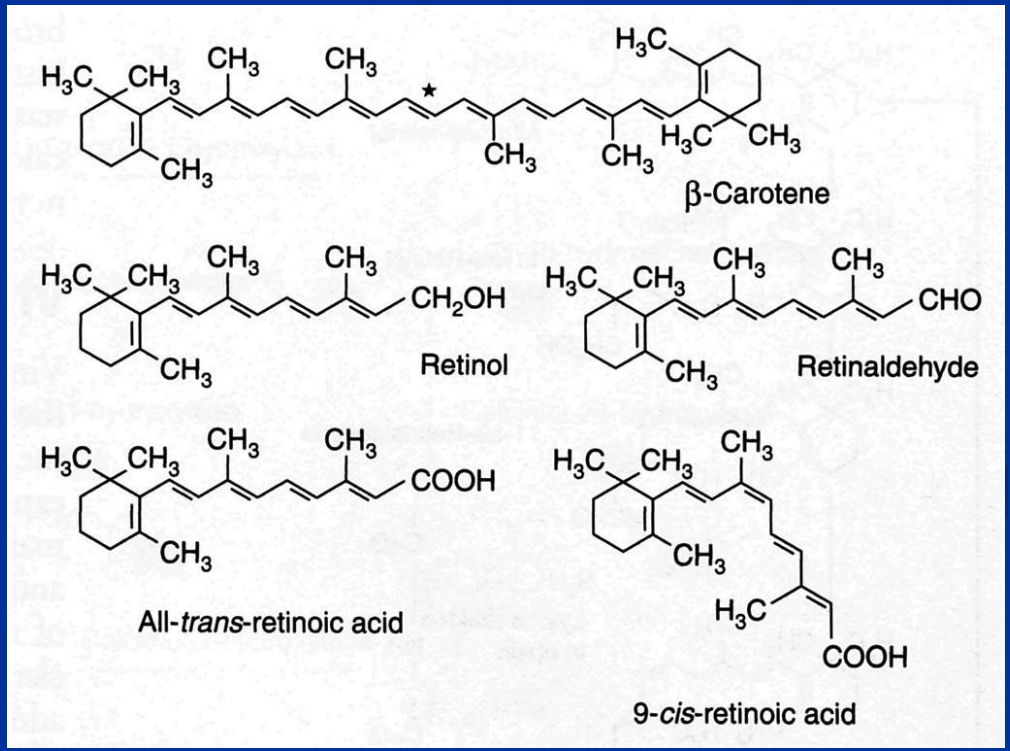
- Deficiency of vitamin A is the most common cause of non-accidental blindness, worldwide
- Preformed
  - Retinoids (retinal, retinol, retinoic acid)
  - Found in animal products
- Provitamin A
  - Carotenoids
  - Must be converted to retinoid form
  - Intestinal cells can split carotene in two (molecules of retinoids)
  - Found in plant products

Lipid-soluble vitamins
Vitamin A

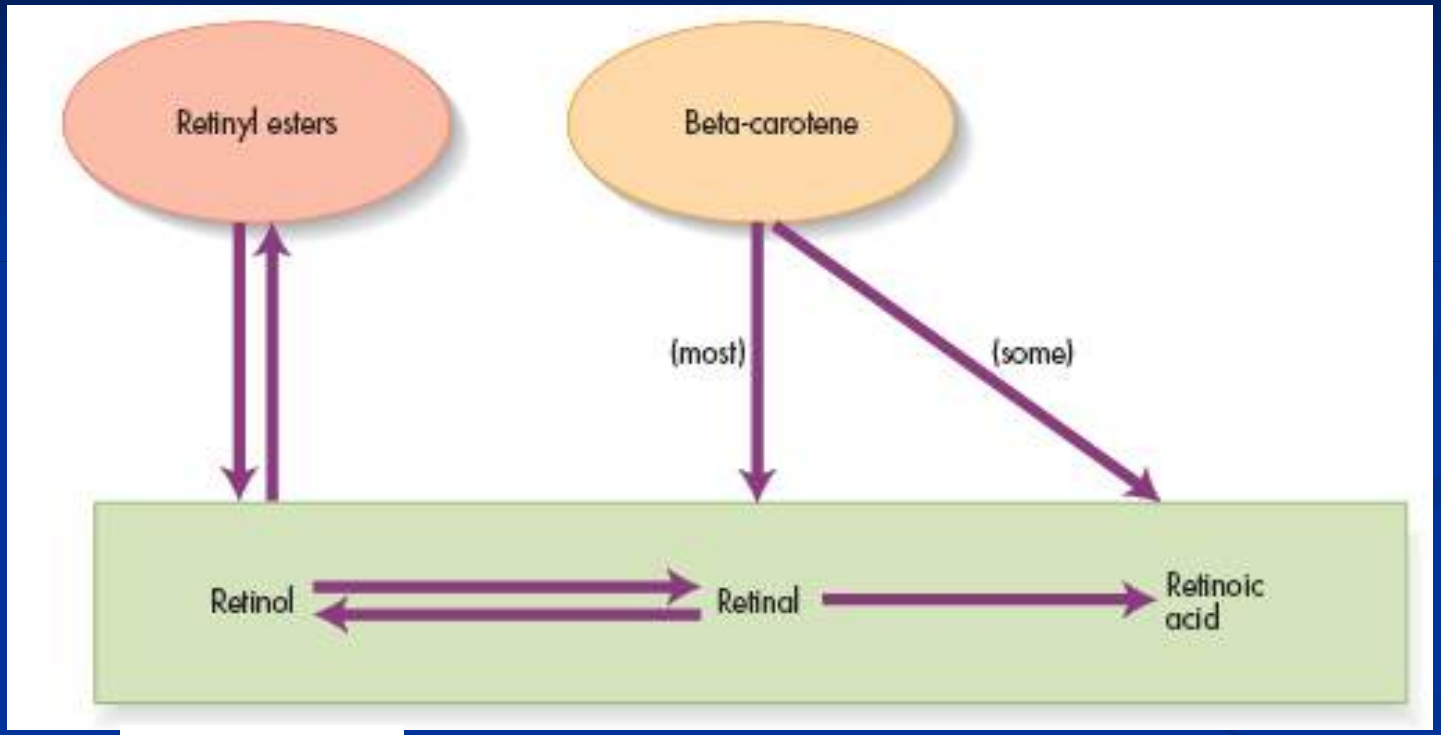
Retinol

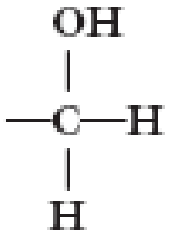
- Biologically active forms - *retinoids*: retinol, retinal, retinoid acid.
- Major vit. A precursors (provitamins) → plants *carotenoids*.
- Foodstaf of animals origin contain most of vit. A in the form of esters (retinylpalmi-tates) – *retinol* and *long fatty acid*

Cyklohexan ring and isoprenoid chain

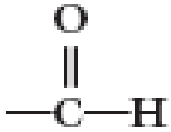


Terminal Ends of Retinoids





Retinol

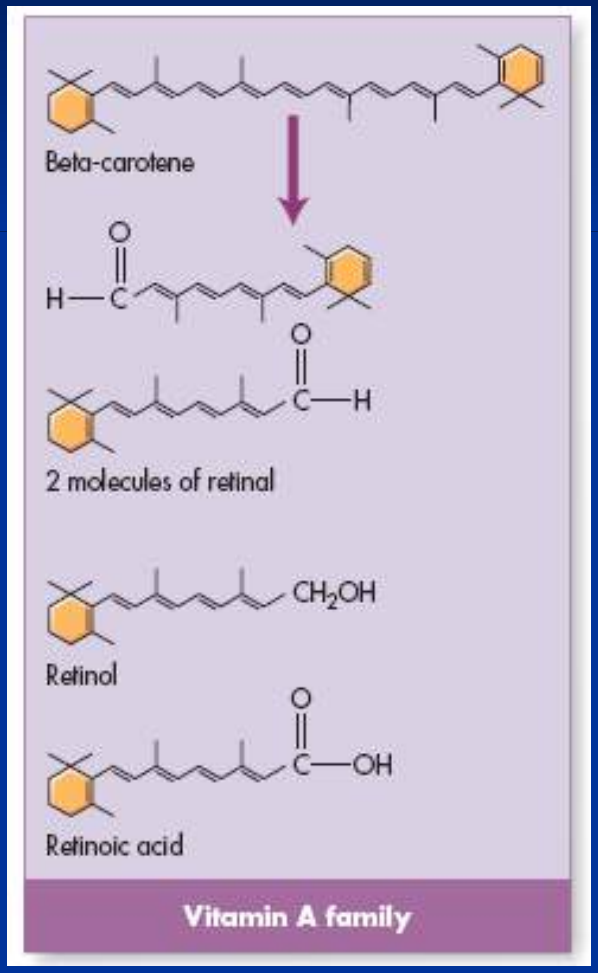


Retinal



Retinoic acid

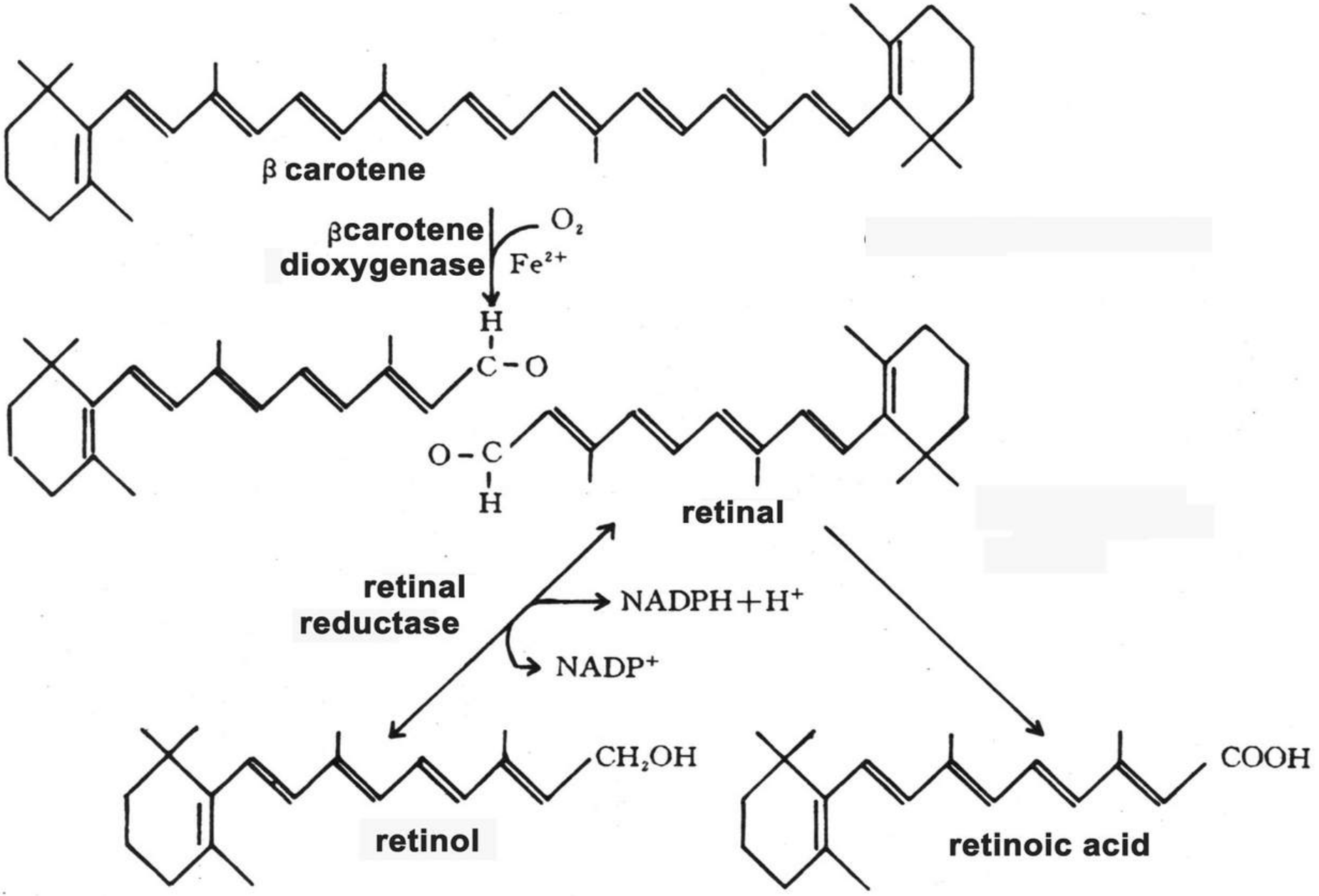
Conversion of Carotenoids to Retinoids



- Enzymatic conversion of carotenoids occurs in liver or intestinal cells, forming retinal and retinoic acid
- Provitamin A carotenoids
  - Beta-carotene
  - Alpha carotene
  - Beta-cryptoxanthin
- Other carotenoids
  - Lutein
  - Lycopene
  - Zeaxanthin

Absorption of Vitamin A

- Retinoids
  - Retinyl esters broken down to free retinol in small intestine - requires bile, digestive enzymes, integration into micelles
  - Once absorbed, retinyl esters reformed in intestinal cells
  - 90% of retinoids can be absorbed
- Carotenoids
  - Absorbed intact, absorption rate much lower
  - Intestinal cells can convert carotenoids to retinoids



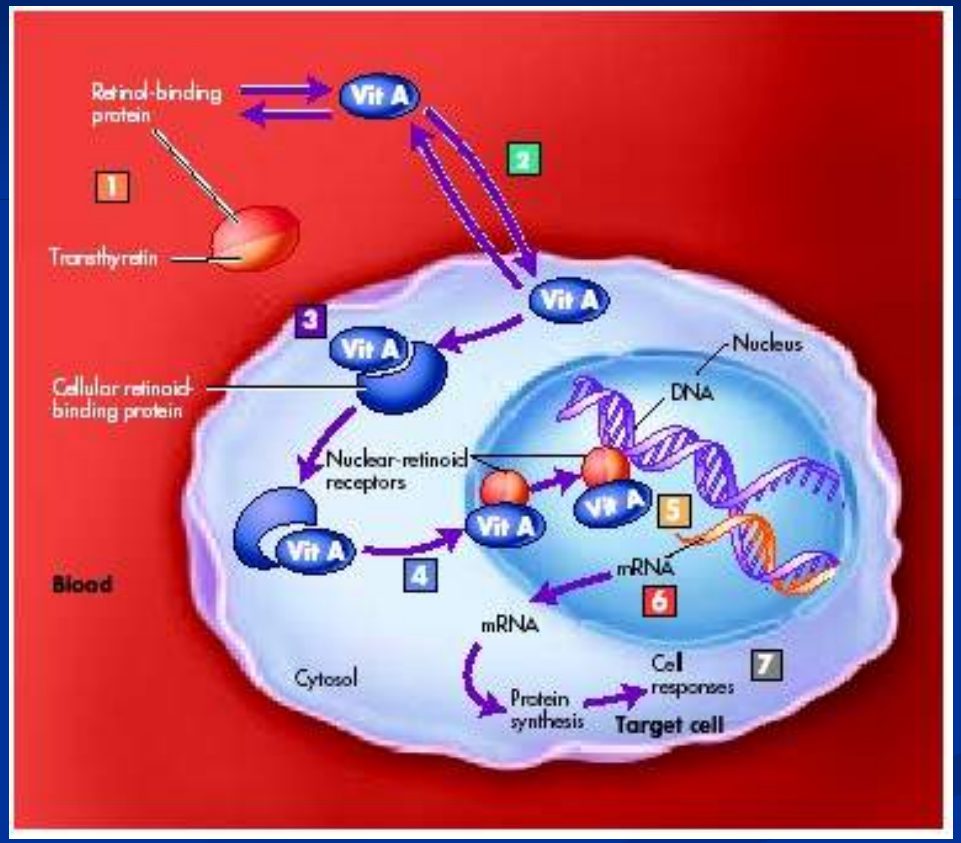
oxidation of carotene to retinal in intestine and conversion to retinol and retinoic acid

Transport and Storage of Vitamin A

- Liver stores 90% of vitamin A in the body
- Reserve is adequate for several months
- Transported via chylomicrons from intestinal cells to the liver
- Transported from the liver to target tissue as retinol via retinol-binding protein.

Retinoid Binding Proteins

- Target cells contain cellular retinoid binding proteins
  - Direct retinoids to functional sites within cells
  - Protect retinoids from degradation
- RAR, RXR receptors on the nucleus
  - Retinoid-receptor complex binds to DNA
  - Directs gene expression



Excretion of Vitamin A

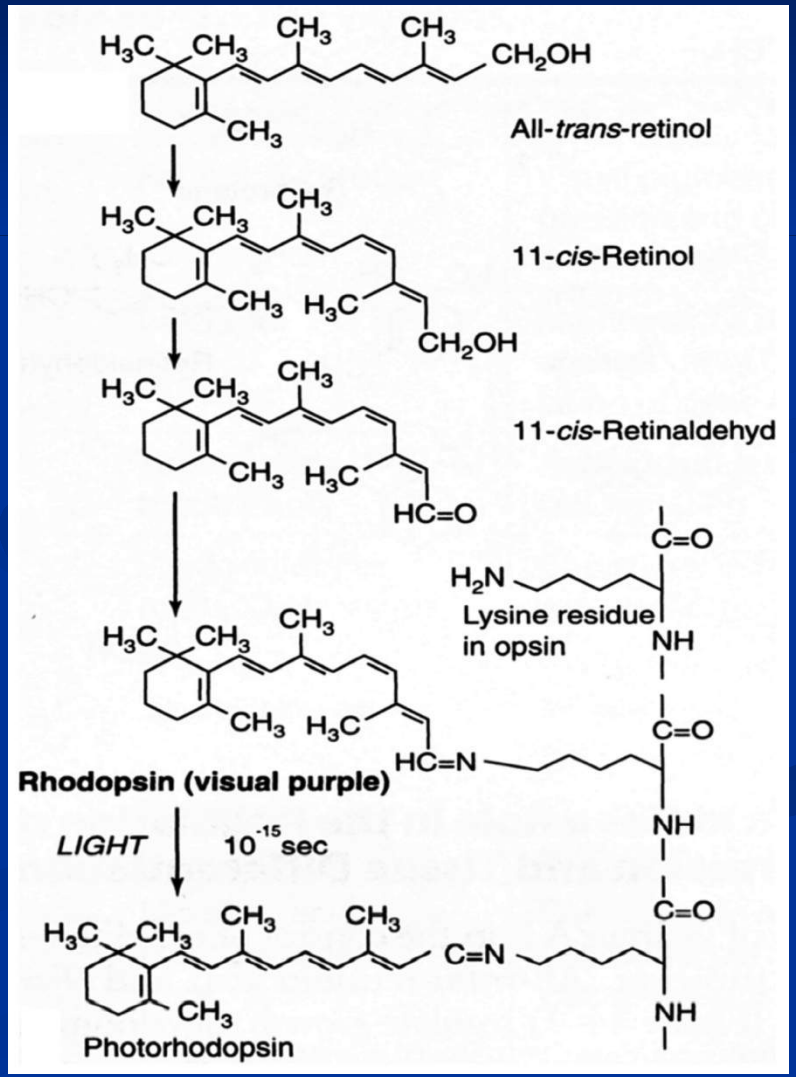
- Not readily excreted
- Some lost in urine
- Kidney disease and aging increase risk of toxicity because excretion is impaired

Functions of Vitamin A: Vision

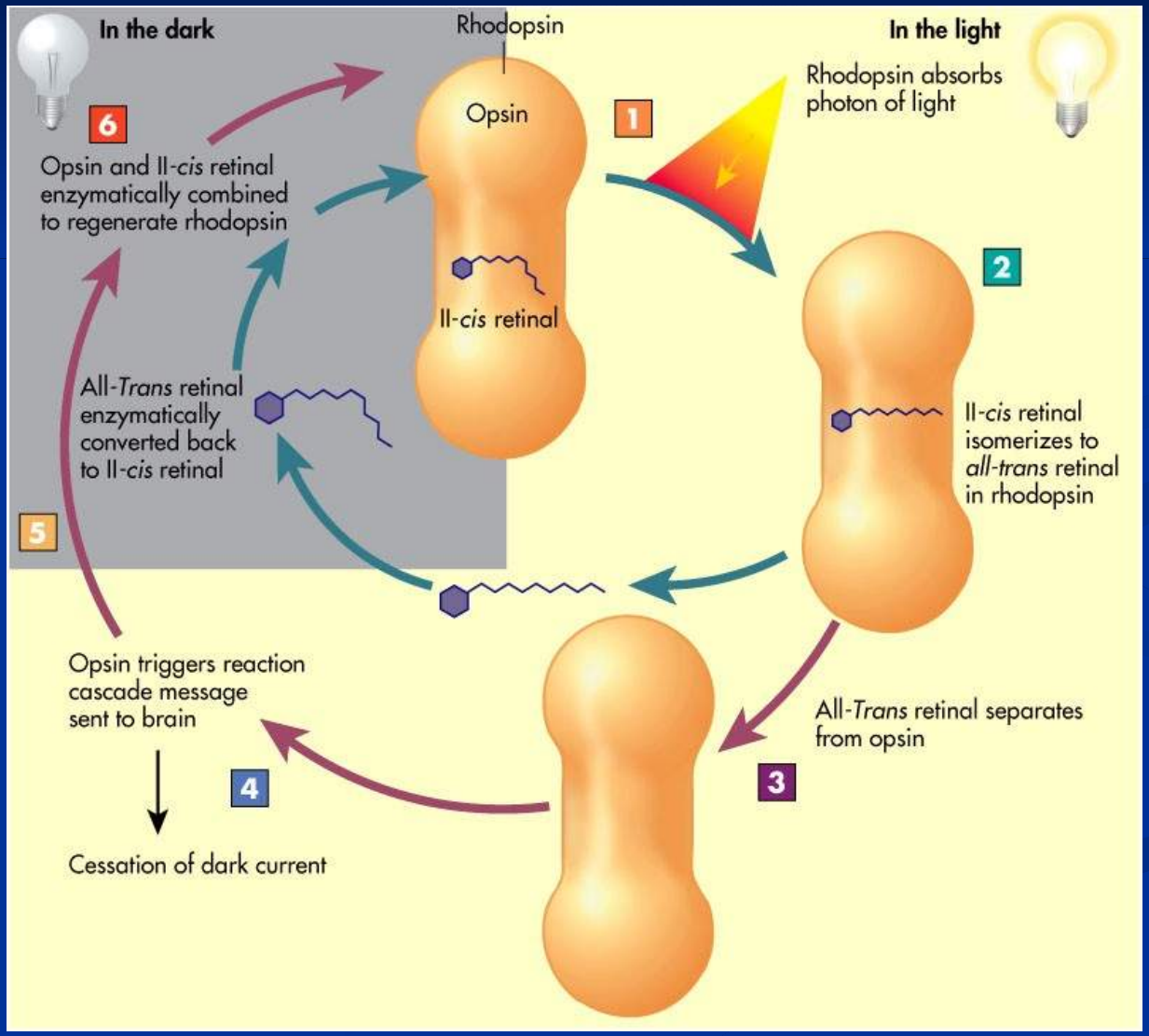
- Retinal turns visual light into nerve signals in retina of eye
- Retinoic acid required for structural components of eye
  - **Cones** in the retina
    - Responsible for vision under bright lights
    - Translate objects to color vision
  - **Rods** in the retina
    - Responsible for vision in dim lights
    - Translate objects to black and white vision

Vitamin A and vision

- Vit. A is necessary to form **rhodopsin** (*in rodes, night vision*) and **iodopsins** (**photopsins**, *in cones – color vision*) - visual pigment.
- Retinaldehyd is a prosthetic group of light-sensitive opsin protein.
- In the retina, all-*trans*-retinol is isomerized to 11-*cis*-retinol → oxidized to 11-*cis*-retinaldehyd, this reacts with opsin (Lys) → to form the holoprotein *rhodopsin*.
- Absorption of light → conformation changes of opsin → photorhodopsin.



The Visual Cycle



Functions of Vitamin A: Growth and Differentiation of Cells

- Retinoic acid is necessary for cellular differentiation
- Important for embryo development, gene expression
- Retinoic acid influences production, structure, and function of epithelial cells that line the outside (skin) and external passages (mucus forming cells) within the body

Vitamin A and other functions

Transcription and cell differentiation

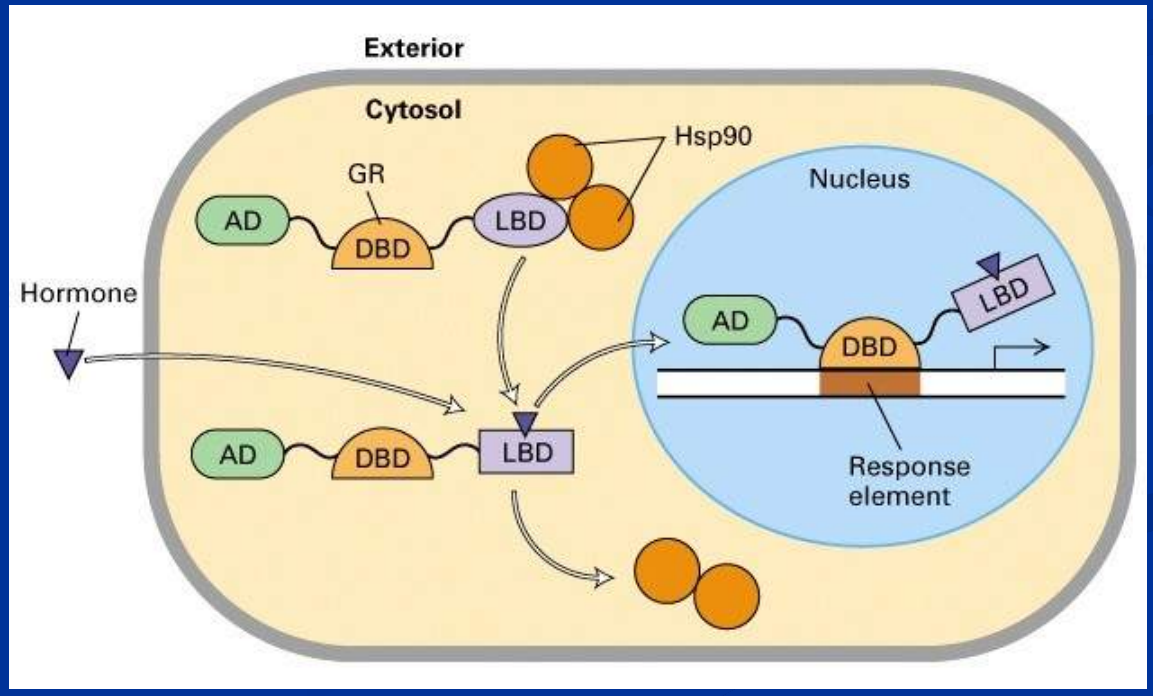
- Retinoic acid regulates the transcription of genes - acts through nuclear receptors (steroid-like receptors).



- By binding to various nuclear receptors, vit. A stimulates (RAR – retinoid acid receptor) or inhibits (RXR- retinoid „X“ receptor) transcription of genes transcription. All-*trans*-retinoic acid binds to RAR and 9-*cis*-retinoic acid binds to RXR.
- Retinoic acid is necessary for the function and maintenance of epithelial tissues.

Nuclear Receptors

Transcription Factors regulated by hydrophobic molecules



Functions of Vitamin A: Immunity

- Deficiency leads to decreased resistance to infections
- Supplementation may decrease severity of infections in deficient person

Vitamin A Analogs for Acne

- Topical treatment (Retin-A)
  - Causes irritation, followed by peeling of skin
  - Antibacterial effects
- Oral treatment
  - Regulates development of skin cells
  - Caution regarding birth defects

Possible Carotenoid Functions

- Prevention of cardiovascular disease
  - Antioxidant capabilities
  - ≥5 servings/day of fruits and vegetables
- Cancer prevention
  - Antioxidant capabilities
  - Lung, oral, and prostate cancers
  - Studies indicate that vitamin A-containing foods are more protective than supplements
- Age-related macular degeneration
- Cataracts
- In general, foods rich in vitamin A and other phytochemicals are advised rather than supplements

Vitamin A in Fo



- Preformed
  - Liver, fish oils, fortified milk, eggs, other fortified foods
  - Contributes ~70% of vitamin A intake for Americans
- Provitamin A carotenoids
  - Dark leafy green, yellow-orange vegetables/fruits

Deficiency of Vitamin A

- Most susceptible populations:
  - Preschool children with low F&V intake
  - Urban poor
  - Older adults
  - Alcoholism
  - Liver disease (limits storage)
  - Fat malabsorption



- Consequences:
  - Night blindness
  - Decreased mucus production
  - Decreased immunity
  - Bacterial invasion of the eye
  - Conjunctival xerosis
  - Bitot’s spots
  - Xerophthalmia
  - Irreversible blindness
  - Follicular hyperkeratosis
  - Poor growth

Upper Level for Vitamin A

- 3000 µg retinol
- Hypervitaminosis A results from long-term supplement use (2 – 4 x RDA)
- Toxicity
- Fatal dose (12 g)

Toxicity of Vitamin A

- Acute – short-term megadose (100 x RDA); symptoms disappear when intake stops
  - Headaches
  - Blurred vision
  - Poor muscle coordination

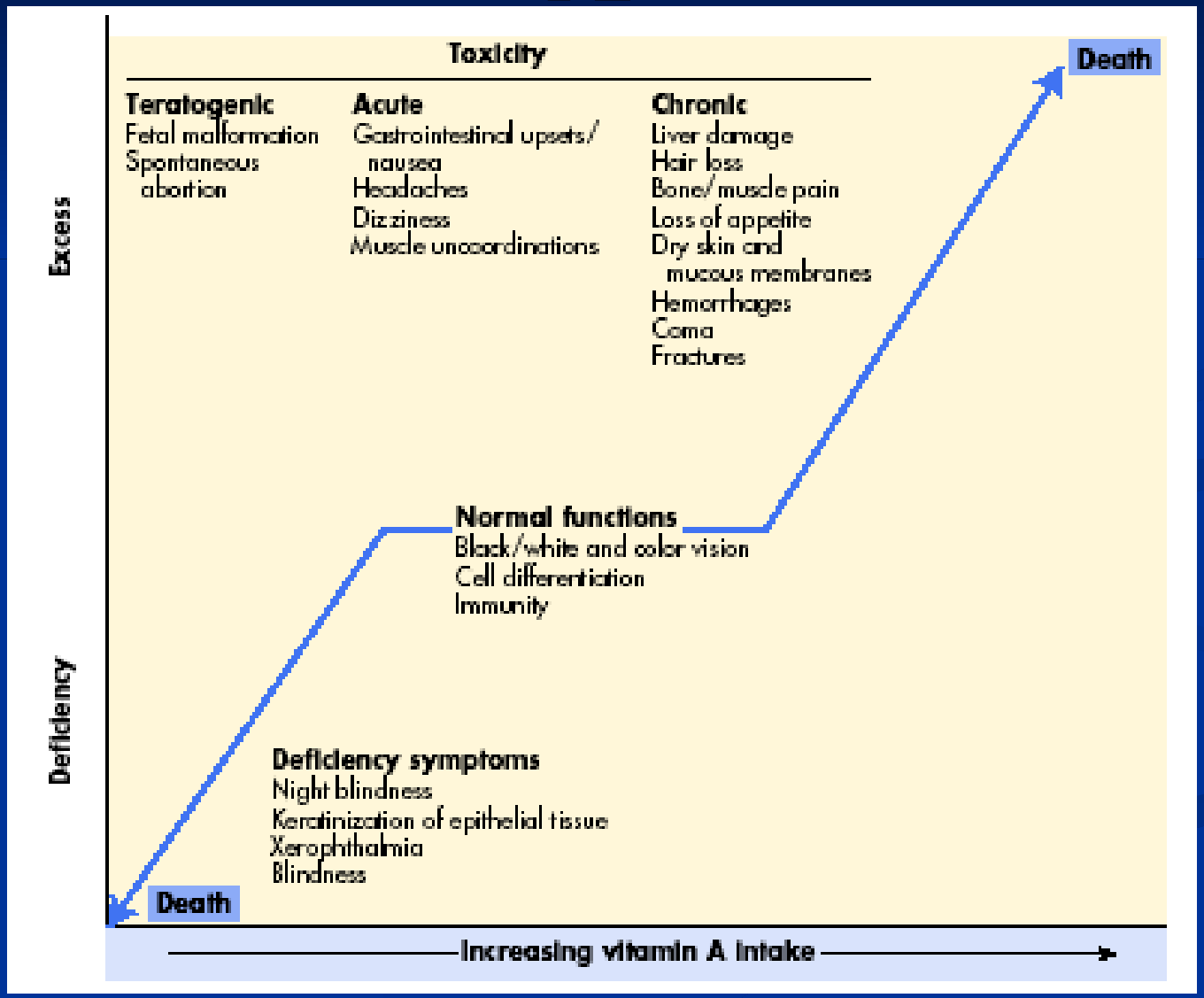
Toxicity of Vitamin A

- Chronic – long-term megadose; possible permanent damage
  - Bone and muscle pain
  - Loss of appetite
  - Skin disorders
  - Headache
  - Dry skin
  - Hair loss
  - Increased liver size
  - Vomiting

Toxicity of Vitamin A

- Teratogenic (may occur with as little as 3 x RDA of preformed vitamin A)
  - Tends to produce physical defect on developing fetus as a result of excess vitamin A intake
  - Spontaneous abortion
  - Birth defects

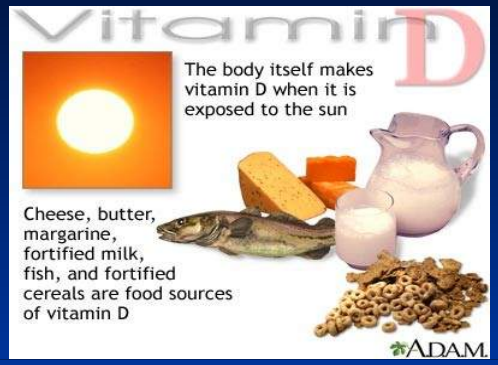
Health Effects of Vitamin A



Toxicity of Carotenoids

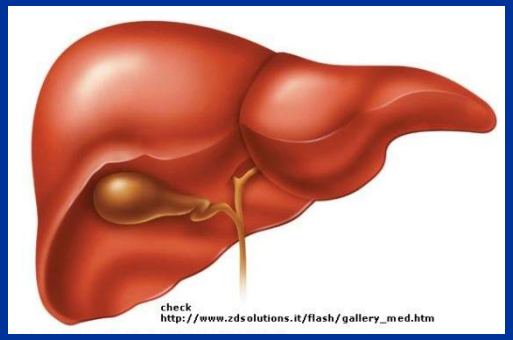
- Not likely, as rate of conversion of carotenoids to retinoic acid by liver is slow and efficiency of absorption of carotenoids decreases as intake increases
- Hypercarotenemia
  - High amounts of carotenoids in the bloodstream
  - Excessive consumption of carrots/squash/beta-carotene supplements
  - Skin turns a yellow-orange color

Vitamin D


Vitamin D
The body itself makes vitamin D when it is exposed to the sun
Cheese, butter, margarine, fortified milk, fish, and fortified cereals are food sources of vitamin D
ADAM.






check
http://www.zdsolutions.it/flash/gallery_med.htm


Vitamin B
Folic acid
Vitamin C
Vitamin D
Vitamin E
Vitamin K
Minerals



Vitamin D

- The term “Vitamin D” refers to a family of related compounds biosynthesized from **cholesterol** and **ergosterol**.

- **Vitamin D3 (cholecalciferol)** is synthesized from **7-dehydrocholesterol** in sun-exposed skin and found naturally in animal products such as **eggs, fish** and **liver**.

- **Vitamin D2 (ergocalciferol)** is another form of vitamin D synthesized by certain **fungi** and is used in many supplements.

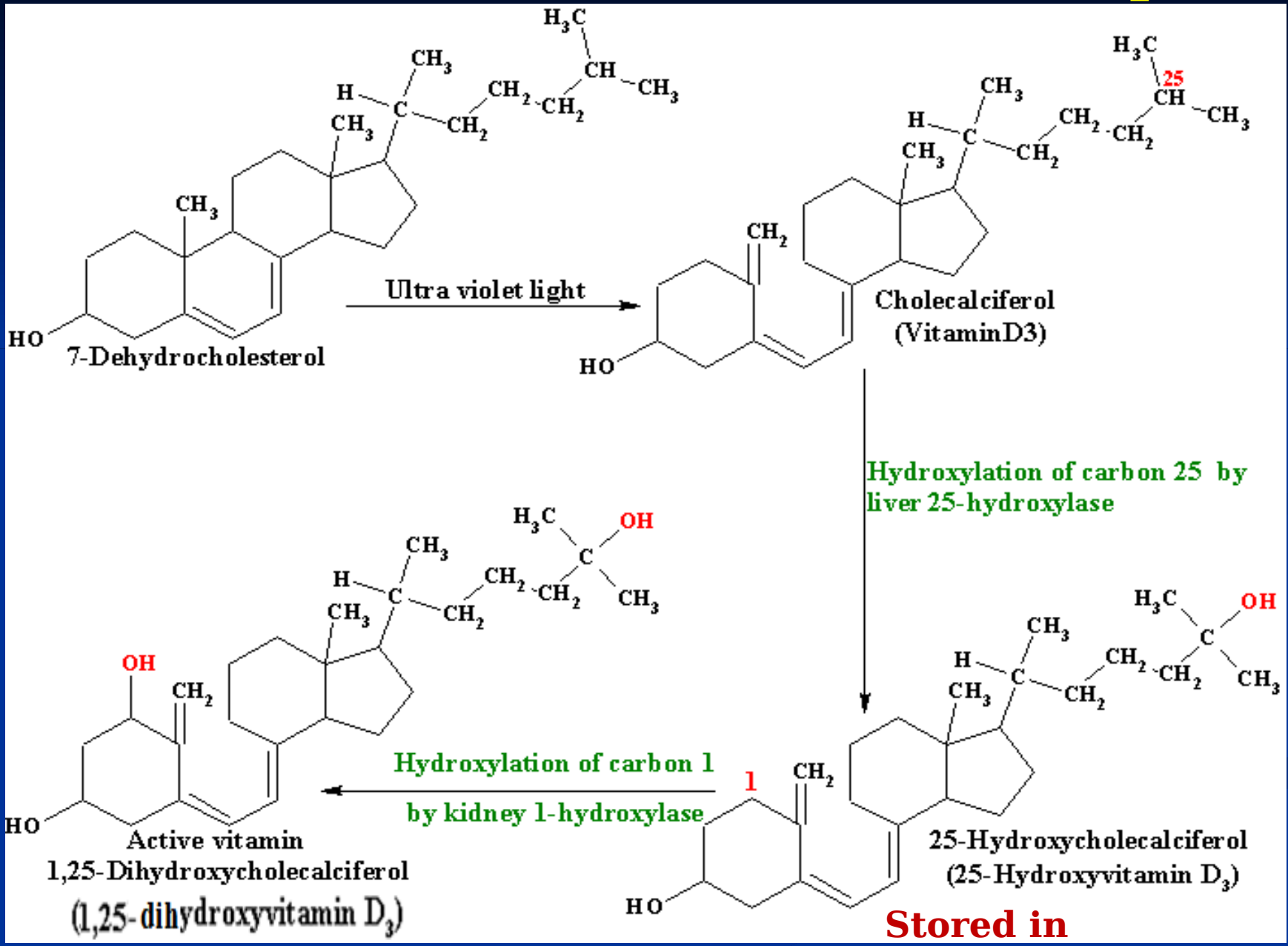
(Cholecalciferol) VITAMIN D_3	(Ergocalciferol) VITAMIN D_2
❖Synthesized in the body	❖Found in plant life
❖Naturally derived supplement	❖Synthetically derived supplement
❖Significantly increases vitamin D levels in the body	❖Moderately increases vitamin D levels in the body
❖Recommended by experts for optimal bone and immune support	❖Alternative form appropriate for vegetarians

- **However, vitamin D3 is the preferred form for humans as its bioavailability is twice that of vitamin D2.**

- **After absorption from foods or production in the skin, vitamin D is stored as 25-hydroxy-vitamin D3 in the liver. When needed by the body, it is then activated by the kidney to 1, 25 - di hydroxy vitamin D3.**

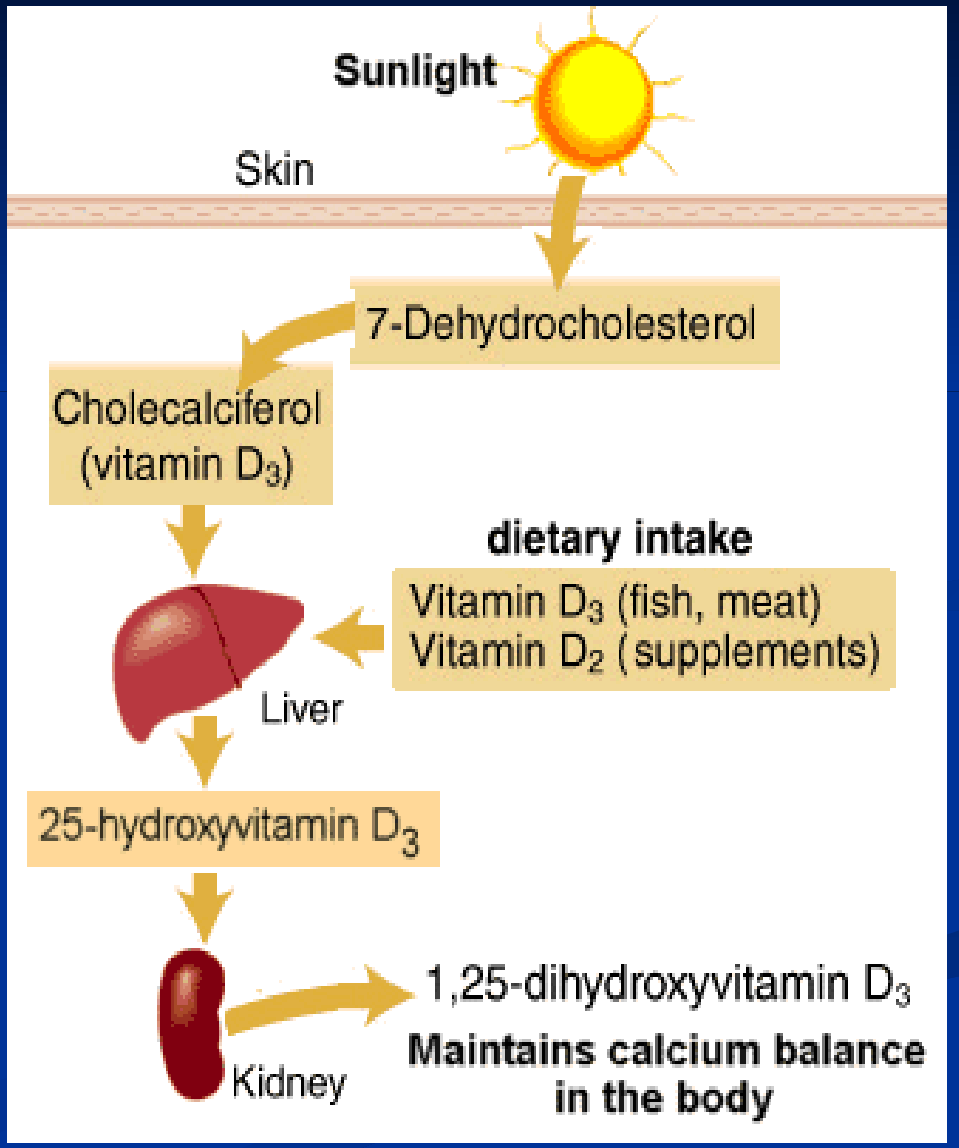
NB: A healthy liver and kidneys are essential for optimum vitamin D status.

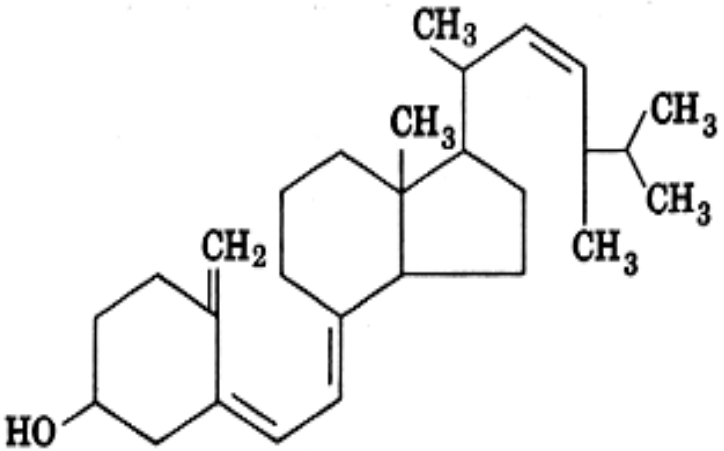
Biosynthesis and Activation of Vitamin D_3



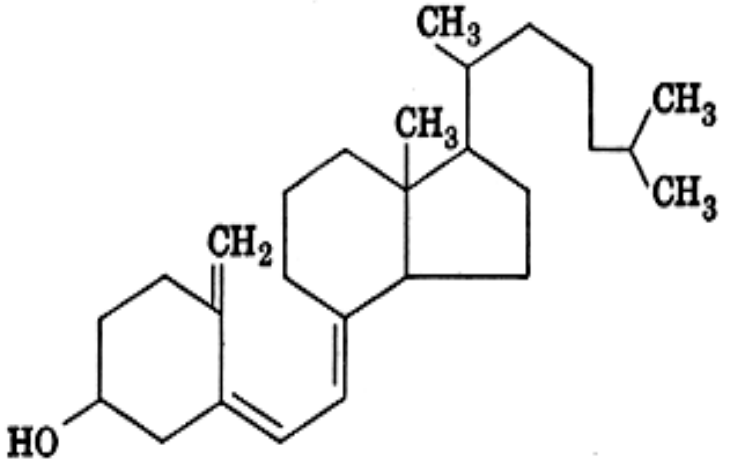
Daily Requirment = 10µg vitamin D = 400 IU vitamin D.



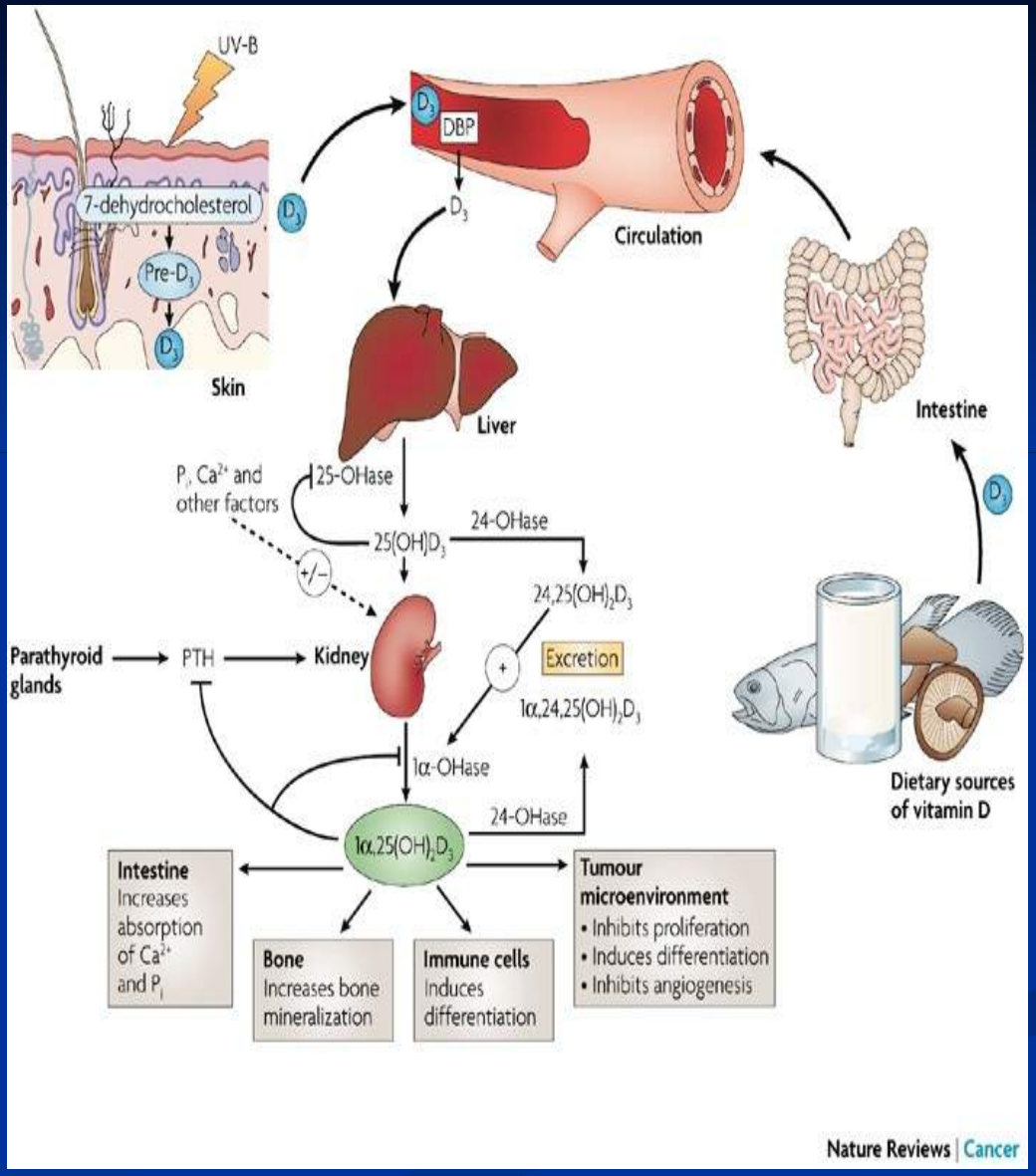




Vitamin D_2 (calciferol)



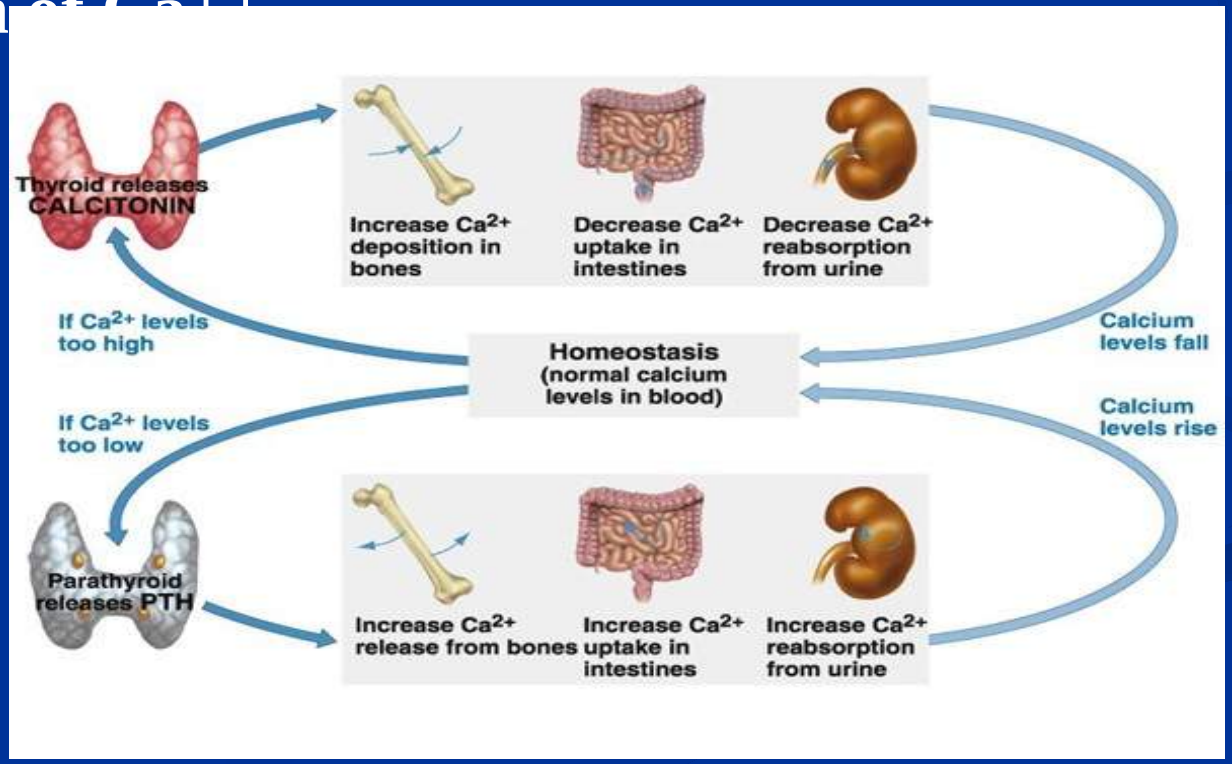
Vitamin D_3



FUNCTIONS OF VITAMIN D

1. Calcium metabolism (main function):

Vitamin D regulates Ca++ levels in the blood and tissues. A fall in blood Ca++ stimulate active vitamin D production stimulates Ca++ absorption from food + release of Ca++ from bones + renal excretion of Ca++



2. Essential for normal bone growth during childhood and for maintaining bone density and strength during adulthood (as it increases Ca++ absorption from foods + increases Ca++ deposition into the skeleton).

3. Integrated function with parathyroid hormone in stabilization of Ca++ level in blood.

4. Regulation of cell growth and development (particularly WBCs and epithelial cells).

PEOPLE AT GREATER RISK OF DEVELOPING VITAMIN D DEFICIENCY

1. Older people (> 50 years) due to:

The decrease of ability of skin to convert 7-dehydrocholesterol to pre-vitamin D3.



2. Newborn infants:

who are absolutely breastfed require vitamin D supplements because mother milk does not contain significant levels of the vitamin.



3. Women after labor: the serum concentration of vitamin D and Ca++ falls rapidly.



5. Dark-skinned individuals may require extra vitamin D because melanin acts like a sun-block, prolonging the time required to generate vitamin D.

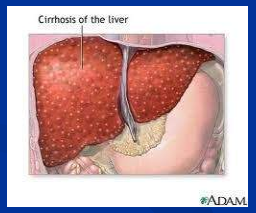


6. Obese people have lower levels of the circulating form of vitamin D, probably because it is deposited in body fat masses

less bioavailable vitamin D.



7. Patients with chronic liver disease or malabsorption or parathyroidectomy.



8. Vitamin D is found only in animal foods.

strict vegetarian people sharply increase the risk of deficiency if sunlight exposure is inadequate.



CAUSES OF VITAMIN D DEFICIENCY

- **Decreased intake or malabsorption.**
- **Reduced sun exposure.**
- **Increased hepatic catabolism (e.g. due to metabolizing induction drug)**
- **Decreased endogenous synthesis (via 25-hydroxylation in the liver and subsequent 1-hydroxylation in the kidney).**

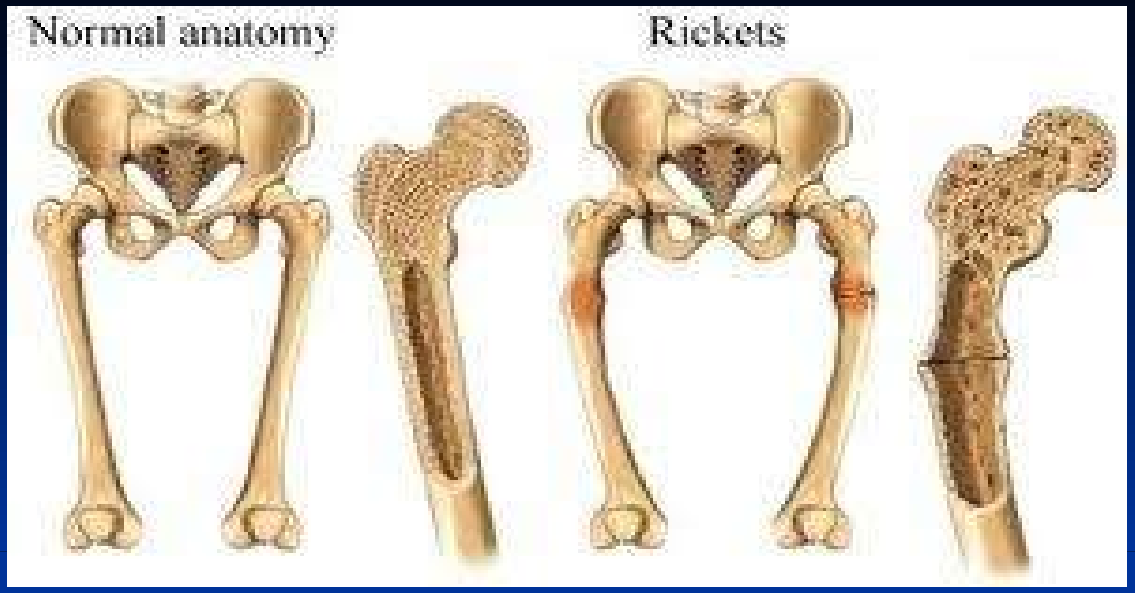
DISEASES CAUSED BY VITAMIN D DEFICIENCY

Vitamin D deficiency causes several bone diseases, including:

Rickets: a childhood disease characterized by failure of growth and deformity of long bones.

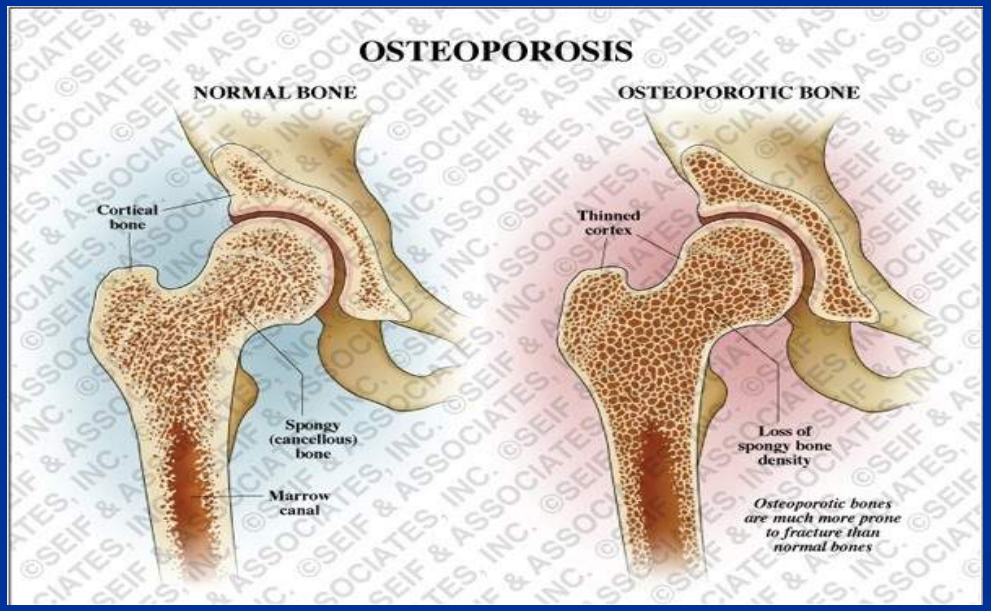
Osteoporosis: a condition characterized by fragile bones due to decreased bone density.
fragile bone easily fractured.

Osteomalacia: (adult version of Rickets) is a case of softening of bones due to defective bone mineralization and characterised by proximal weakness and bone fragility.

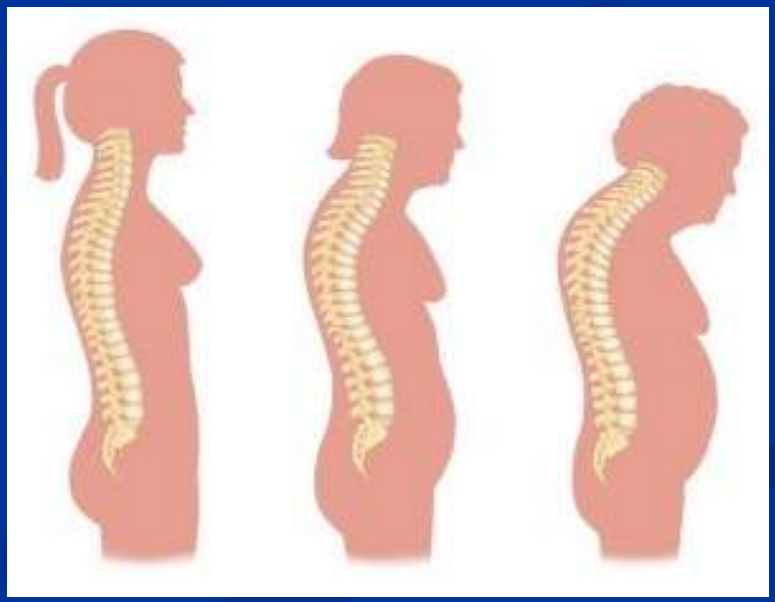


Rickets





Osteoporosis



Osteomalacia

GENERAL USES OF VITAMIN D

1- **Prophylaxis and treatment of osteoporosis** as a result of calcium depletion. Treatment require taking both calcium and vitamin D.

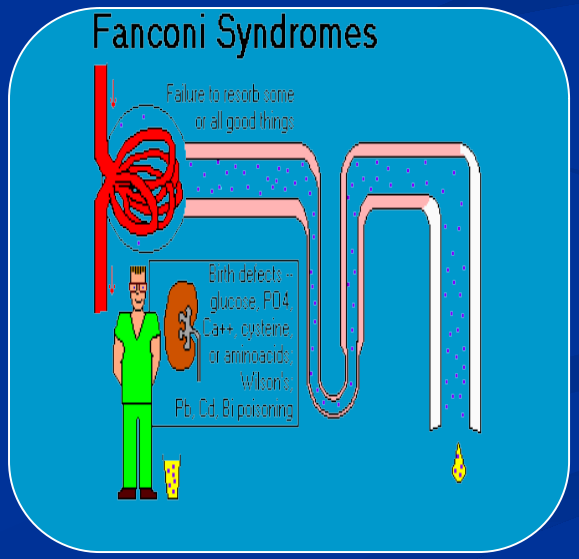
2- **Trteatment of Osteomalacia and rickets.**

3- **Vitamin D also has a role in cancer prevention** (colon cancer, breast and prostate cancer). The action of Tamoxifen (Chemotherapeutic agent) appears to be improved with small added doses of vitamin D.

4- **Treatment of hearing loss** resulted from vitamin D deficiency that affect the function of small bones in the ear responsible sound transmition.

6- In combination with phosphate to treat some metabolic diseases as Fanconi syndrome and familial hypophosphatemia.

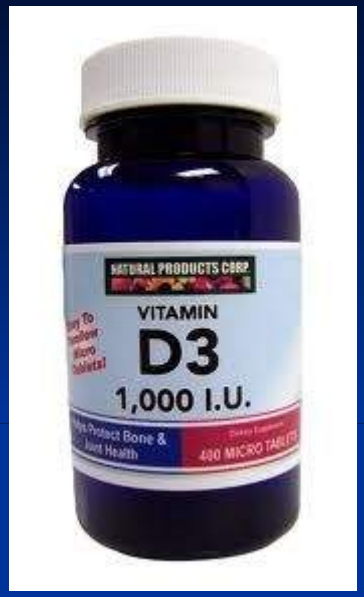
Fanconi syndrome: is a disorder in which the proximal tubular function of the kidney is impaired, resulting in decreased re- absorption of electrolytes and nutrients back into the bloodstream (compounds involved include glucose, amino acids, uric acid, phosphate and bicarbonate).



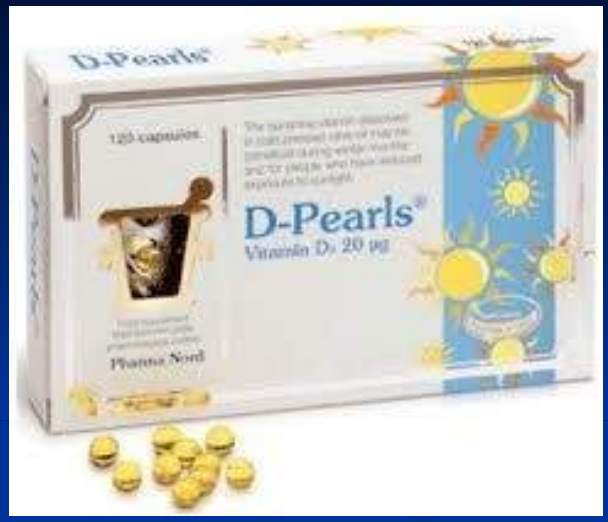
INTERACTIONS WITH VITAMIN D

1. **The absorption of vitamin D is improved** by calcium, choline, fats, phosphorus, vitamins A and C.
2. **Drugs as Rifampin, H2 blockers, barbiturates, heparin, cholestyramine, carbamazepine, phenytoin, fosphenytoin, and phenobarbital** reduce serum levels of vitamin D and increase its metabolism.
3. **Overuse of mineral oil** and stimulant laxatives may **deplete vitamin D**.
4. Osteoporosis and hypocalcaemia can result from prolonged use of corticosteroids. It is necessary to take of **calcium and vitamin D together with corticosteroid drugs.**

DOSAGE FORMS













Fat Soluble Vitamins

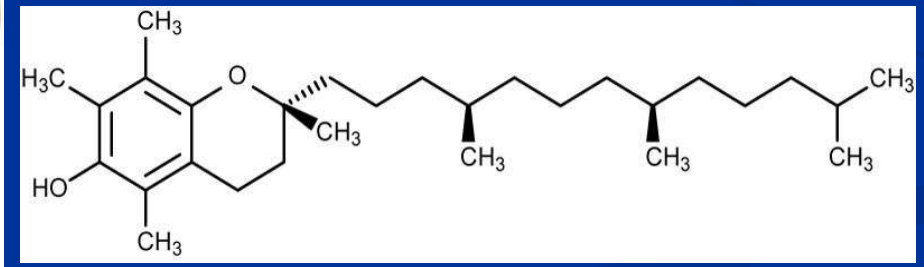


VITAMIN E





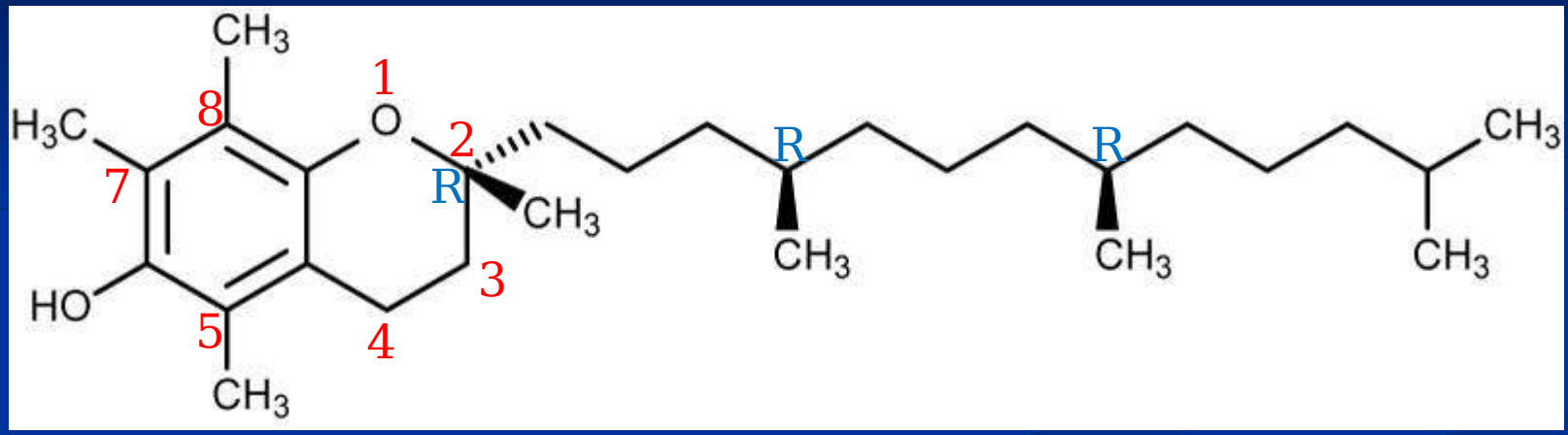






Vitamin E

- **Vitamin E is a general term used for a group of compounds (α, β, γ, and δ-tocopherols) having a chromanol ring and phytyl side chain.**



Position of Methyl groups on aromatic ring	Tocopherol structure
5,7,8	α-tocopherol
5,8	β-tocopherol
7,8	γ-tocopherol
8	δ-tocopherol

d -α -tocopherol Equivalent (TE) = 1 mg = 1.5 IU
Recommended daily amount = 30 IU = 20 mg
Larger doses are allowed (safe)

- The most abundant and active form of vitamin E is α-tocopherol (α-*RRR-tocopherol).*
- The relative activities of the tocopherols vary considerably and thus vitamin E activity should referred to α-tocopherol.

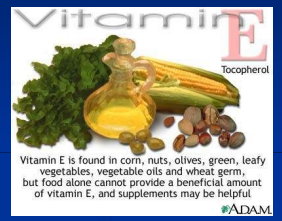
Vitamin E form	Relative Biological Activity
α-tocopherol	100%
β-tocopherol	50%
γ-tocopherol	10-30%
δ-tocopherol	1%

- F… content than sunflower oil. Most of the vitamin E in soya oil is δ-tocopherol while in sunflower oil most of the vitamin E is α-tocopherol, which gives the sunflower oil a greater level of vitamin E activity.

Sources of Vitamin E

- Natural sources of vitamin E (d-RRR-α-tocopherol)
  - Vegetable oils from seeds (Sunflower, Cottonseed)
  - Lettuce.
  - Nuts.
  - Unprocessed grains.



- Synthetic sources ⟶ racemic-tocopherols.

Pharmacokinetics of Vitamin E

- Gastrointestinal absorption of all forms of vitamin E is equivalent.
  - The subsequent biological steps are sharply in favour of the RRR form.
  - The cellular liver transfer protein that maintains the plasma level is specific for the RRR form of α-tocopherol.

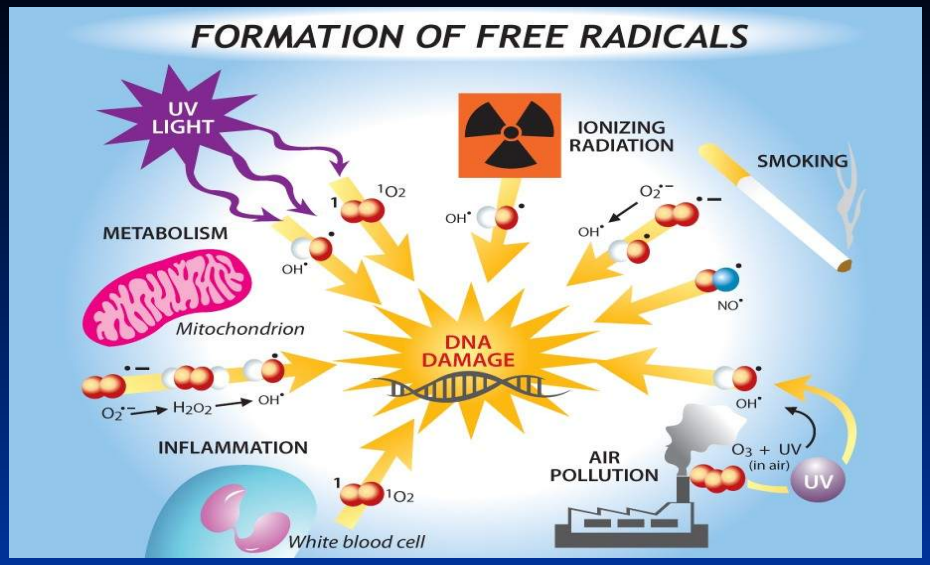
Functions of Vitamin E

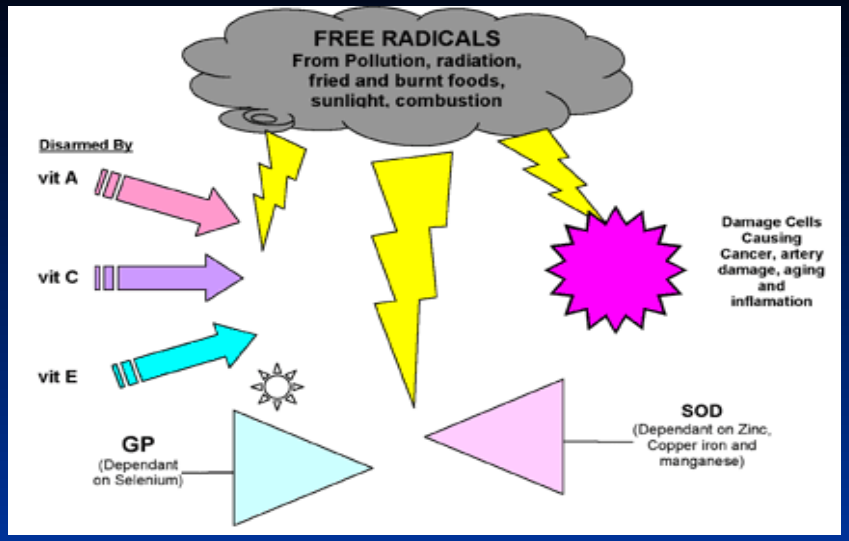
1. **Antioxidant action:**

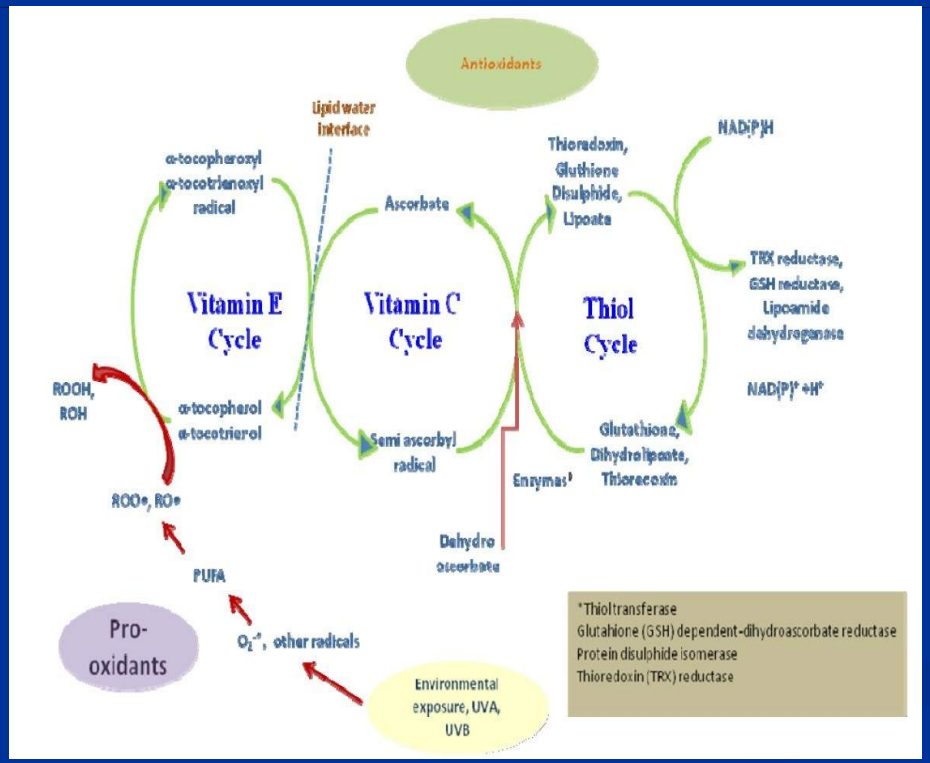
- Tocopherols (Vitamin E) interrupt free radical chain reactions by capturing the free radical (free radical scavanger); this imparts to them their antioxidant properties.

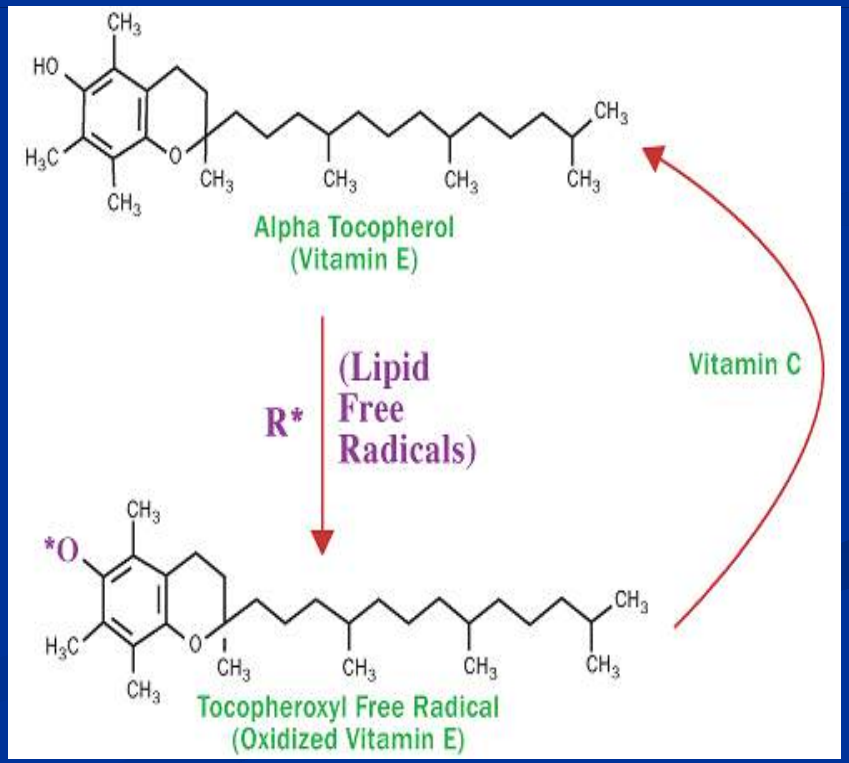
- Therefore, vitamin E protects the cells, with other nutrient (e.g. Vitamin C and Selenium) against the harmful free radicals formed during metabolism of fatty acids.

- The free hydroxyl group on the aromatic ring is responsible for the antioxidant properties. The hydrogen from this group is easily donated to the free radical, resulting in a relatively stable free radical form of the vitamin.


FORMATION OF FREE RADICALS
UV LIGHT
IONIZING RADIATION
SMOKING
METABOLISM
Mitochondrion
DNA DAMAGE
INFLAMMATION
AIR POLLUTION
White blood cell
UV


FREE RADICALS
From Pollution, radiation, fried and burnt foods, sunlight, combustion
Disarmed By
vit A
vit C
vit E
GP
(Dependant on Selenium)
SOD
(Dependant on Zinc, Copper iron and manganese)
Damage Cells Causing Cancer, artery damage, aging and inflamation


Antioxidants
Lipid water interface
Vitamin E Cycle
Vitamin C Cycle
Thiol Cycle
Ascorbate
Semi ascorbyl radical
Dehydro ascorbate
Thioredoxin, Gluthione Disulphide, Lipoate
NADPH
TRX reductase, GSH reductase, Lipoamide dehydrogenase
Glutathione, Dihydrolipoate, Thioredoxin
Enzymes*
ROOH, ROH
ROO•, RO•
PUFA
Pro-oxidants
Environmental exposure, UVA, UVB
*Thioltransferase
Glutathione (GSH) dependent-dihydroascorbate reductase
Protein disulphide isomerase
Thioredoxin (TRX) reductase


Alpha Tocopherol
(Vitamin E)
R*
(Lipid Free Radicals)
Vitamin C
Tocopheroxyl Free Radical
(Oxidized Vitamin E)

2. Antithrombotic action:

Vitamin E slows down the action of thrombin (a blood clotting protein) and reduces platelet aggregation by inhibiting thromboxane. Although vitamin E is a natural “blood thinner” it does not increase risk of bleeding in healthy people.

Uses of Vitamin E as prophylaxis and Therapy

1. **Cardiovascular disease:**

(e.g. angina pectoris and venous thrombosis): It ↓ oxidation of cholesterol in LDL and VLDL particles → ↓ risk of coronary heart disease.

2. **Anemia:**

It functions as an antioxidant, enhancing function and durability of RBCs and stabilizes their membranes → ↓ syndromes of hemolytic and sickle-cell anemia.

3. **Immunity:**

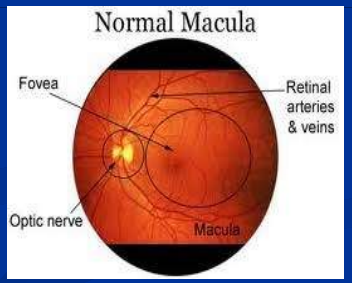
It ↑ resistance to viral and bacterial infections and ↑ antibody production by WBCs and ↑ phagocytosis.

4. Alzheimer's disease (AD) and memory loss:

It can ↓ progression of AD and help maintain memory function through decreasing oxidative damage to neurons.

5. Eye Diseases:

It ↓ oxidative damage to the lens, ↓ developing of cataract. It also ↓ incidence of ↓ macular degeneration.



6. Rheumatic disorders and pains:

It acts as free radical scavenger and so as anti-inflammatory and analgesic. It can ↓ joint inflammation and stiffness in osteoarthritis and rheumatoid arthritis.

7. Protection and treatment of cancer:

Higher intakes of vitamin E helps ↓ risk of cancer of the skin, breast, lung, esophagus, and stomach. It can also ↑ efficacy of the radiation treatment and protects healthy cells against damage during radiotherapy.

8. Skin Care:

It is applied topically to abrasions, wounds, or burns ⟶ reduce scar formation and contraction, and improve healing.

9. Hepatoprotective

10. Disorders of premature infants:

It ↓ risk of hemolytic anemia and bronchopulmonary dysplasia (characterized by inflammation and scarring in the lungs).

Toxicity and side effects

- **Vitamin E is well-tolerated**, and side effects are rare even at doses of (up to 2000 mg/day).

- However, in some individuals who are vitamin K deficient, vitamin E may increase the risk for hemorrhage or bleeding.

- People taking anticoagulant drugs should also be cautious with high doses.

- People with diabetes should be cautious when starting high doses of vitamin E because it may enhance the action of insulin but, rarely.

People at greater risk of developing vitamin E deficiency

- **Many people** do not obtain adequate dietary vitamin E **when depends more on processed food** (e.g. refined flour and white rice lose nearly all of original vitamin present in whole grain).

- **People who consume much of polyunsaturated fatty acids** (PUFAs) as in USA and Europe need more vitamin E to protect PUFAs in their tissues from oxidation particularly if dietary intake of vitamin E is marginal.

- **Newborn and premature infants** are at high risk for deficiency due to their poor ability to absorb vitamin E and very limited tissue reserves of vitamin E.

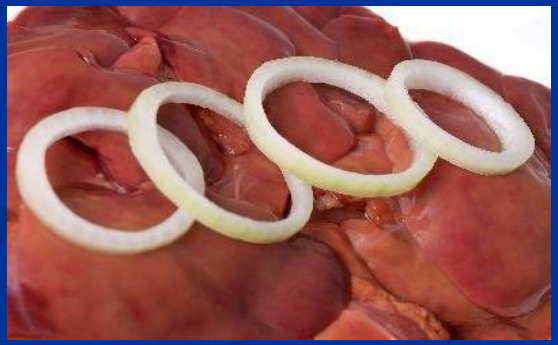
Drug and Herbal Interactions with Vitamin E

- Vitamin E can alter the efficacy of anticoagulant.
- Vitamin E increase bleeding time in patient taking herbal preparations as Feverfew or *Gingko biloba*.
- Inorganic iron supplements destroy vitamin E, so individuals taking iron supplements should space out their doses (e.g., iron in the morning and vitamin E in the evening).
- Large doses of vitamin A can decrease the absorption of vitamin E.
- Alcohol and mineral oil can also reduce vitamin E absorption, and these substances should be avoided in vitamin E deficient individuals.


Vitamin K
Food sources of vitamin K include cabbage, cauliflower, spinach and other green, leafy vegetables, as well as cereals

VITAMIN K



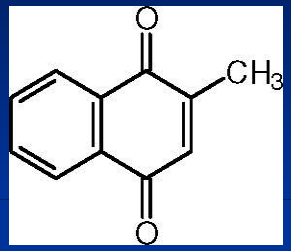






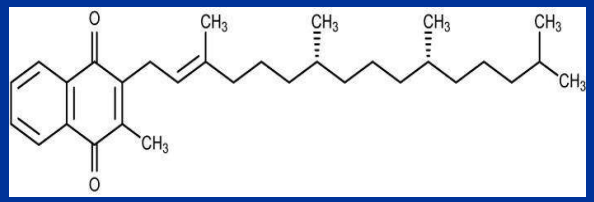
VITAMIN K

❖A group of compounds derived from 2-methyl-1,4-naphthoquinone that act as antihemorrhagic factor in birds and mammals. There are three forms of vitamin K:

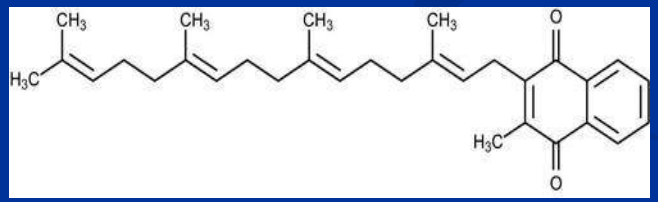


2-methyl-1,4-naphthoquinone

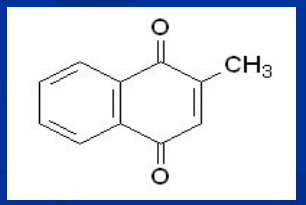
a) Vitamin K1 (Phylloquinone) found in plant foods.
b) Vitamin K2 (Menaquinone) from animal and bacterial sources.
c) Synthetic Vitamin K3 (Menadione).



Vit. K_1 (**Phylloquinone**)



Vit. K_2 (**Menaquinone**)



Vit. K_3 (**Menadione**)

Commercial infant formula contain 50- 125 µg/day

Recommended Daily amount →
New born: 500- 1000 µg once
1- 6 Months: 5 mg/day
6- 12 Months: 15 mg/day
Adult: 80 mg/Kg (body weight) / day

Sources of Vitamin K

Foods rich in Vitamin K:

Spinach, Green cabbage, Turnip, Parsley, lettuce, beef liver, green tea (in decreasing order) etc.



Spinach



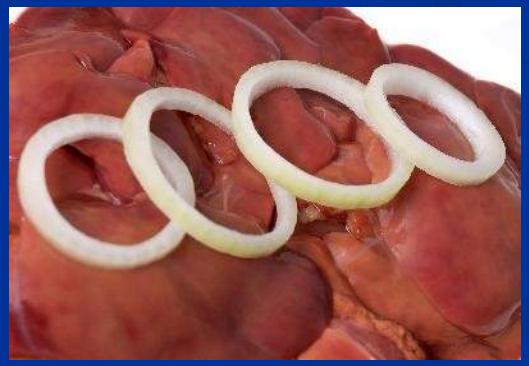
Green cabbage



Turnip



Parsley



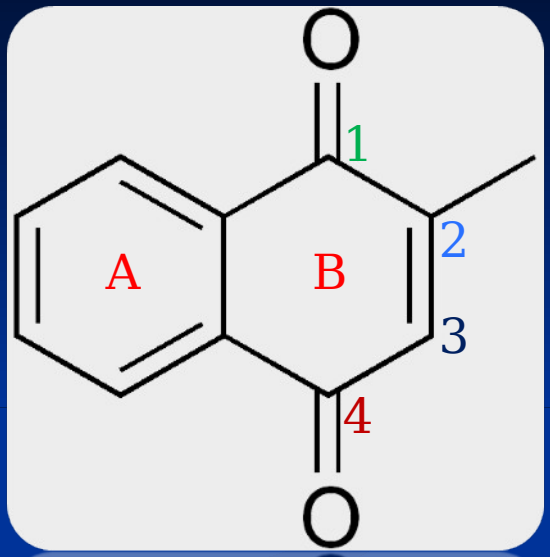
Beef liver



Green tea

Structure Activity Relationship

1. **Activity is maximum when:**



- Ring A and Ring B are aromatic.
- Ring A is not substituted.
- Methyl group at C-2.
- Unsaturation of phytyl side chain at C-3.

2. **Activity is decreased when:**

- Alkyl group larger than Methyl at C-2.
- Hydroxyl group at C-3.
- Hydroxylation or saturation or cis-configuration of the phytyl side chain at C-3.

Activity not affected:

if positions 1 and 4 are substituted with OH, O-Ac, NH_2, OCH_3, OC_2H_5, C=O

Metabolism of Vitamin K

1. 70% of Vit. K3 (menadione) excreted in urine in 24 hr in the form of sulphate, phosphate and glucuronic acid conjugate.
2. Small % excreted in faces as glucuronic acid conjugate.
3. Vit K2 and K3 undergo slower metabolism to shorten the side chain to 5- 7 carbons carboxylate.

Symptoms of Vitamin K deficiency

1. Uncontrolled internal bleeding.
2. Cartilage calcification and malformation of developing bone.
3. Deposition of insoluble calcium salts in the arterial vessel walls.

The basic factors needed to prevent vitamin K deficiency:

a) A normal diet containing the vitamin.
b) The presence of bile in the intestine.
c) A normal intestinal uptake or absorption.
d) A normal liver i.e. no interference with vitamin K metabolism or use of vitamin K antagonists therapeutically or accidently.
e) Heavy alcohol consumption impairs the liver's ability to produce vitamin K-dependent coagulation factors and impair recycling of vitamin K.

People at greater risk of developing Vitamin K deficiency

- Newborn infants who are exclusively breastfed are susceptible to abnormal bleeding due to vitamin K deficiency.
- Breast milk contains very little vitamin K and the immature liver of the newborn does not synthesize the vitamin K-dependent clotting factors efficiently.
- Also, because the newborn's colon is sterile for the first few days after birth, no bacterial synthesis of vitamin K occurs in the colon.
- To reduce the risk of vitamin K deficiency, most babies receive intramuscular vitamin K at birth.

Symptoms of Vitamin K Toxicity

- Vitamins K1 and K2 are non-toxic in large doses.
- Vitamin K3 (the synthetic one) is toxic in doses three time more than the usual dose.

Toxicity manifestations:

- Hyperbilirubinemia.
- Severe Jaundice.
- Anemia.

Drug Interaction with Vitamin K

A. Some interactions may increase the need for vitamin K:

1. **Antibiotics:** Prevent absorption and kill normal bacterial folra.
2. **Anticonvulsants** e.g. Phenytoin : Affect Vit K metabolism.

Functions of Vitamin K

Vitamin K is involved as a cofactor in the carboxylation of certain glutamate residues in proteins to form **γ- carboxyglutamate residues (Gla-residues).** Gla-residues are usually involved in binding calcium and are essential for the following biological activities:

1. **Blood coagulation**

   Production of proteins that are part of the coagulation cascade in the blood. Several proteins promote coagulation (prothrombin, VII, IX, X) while others slow it down (proteins C and S). Thus, activity of vitamin K balances the two opposing sides of coagulation system in blood.

2. **Bone metabolism**

   •Bone Gla-protein (Osteocalcin): Regulate incorporation of calcium Phosphate into bones.

   •Matrix GLA protein (MGP): Clearance of extracellular Calcium to protect against soft tissue calcification.

3. Bile acid sequestrants (Cholestyramine) and Aspirin: Affect absorption.

4. **Weight Loss Products** (Chitosan, Orlistat, and olestra): Affect absorption as they prevent absorption of fats.

5. **Mineral oil laxatives**: Affect absorption.

6. **X-rays and Radiation**: Deplete vitamin K levels and raise vitamin K requirements.

THANX